2. Occurrence
Maximum synthesis occurs during climacteric ripening of fruits (*see* Box info) and tissues undergoing senescence. It is formed in almost all plant parts like roots, leaves, flowers, fruits and seeds.

3. Transport in plants
Ethylene can easily diffuse inside the plant through intercellular spaces.

4. Precursor
It is a derivative of amino acid methionine, linolenic acid and fumaric acid.

5. Bioassay (Gas Chromatography)
Ethylene can be measured by gas chromatography. This technique helps in the detection of exact amount of ethylene from different plant tissues like lemon and orange.

6. Physiological Effects
- Ethylene stimulates respiration and ripening in fruits.
- It breaks the dormancy of buds, seeds and storage organs.
- It stimulates formation of abscission zone in leaves, flowers and fruits. This makes the leaves to shed prematurely.
- Inhibition of stem elongation (shortening the internode).
- Growth of lateral roots and root hairs. This increases the absorption surface of the plant roots.
- Ethylene normally reduces flowering in plants except in Pine apple and Mango.

15.2.5 Abscisic Acid (ABA) (Stress Phyto hormone)

1. Discovery
In 1963, the hormone was first isolated by **Addicott *et al.,*** from young cotton bolls and named as **Abscission II**. Eagles and Wareing during 1963–64 isolated a dormancy inducing substance from leaves of *Betula* and called it as dormin. In 1965, it was found by Cornsforth *et al.*, that both dormin and abscission are chemically same compounds and called **Abscisic Acid (ABA)**.

2. Occurrence
This hormone is found abundantly inside the chloroplast of green cells.

3. Precursors
The hormone is formed from mevalonic acid pathway or xanthophylls.

4. Transport in plants
Abscisic acid is transported to all parts of the plant through diffusion as well as through phloem and xylem.

5. Chemical structure
It has carotenoid structure.

6. Bioassay (Rice Coleoptile)
The inhibition of IAA induces straight growth of rice seedling coleoptiles.

7. Physiological effects
- It helps in reducing transpiration rate by closing stomata.
- ABA is a powerful growth inhibitor. It causes 50% inhibition of growth in Oat coleoptile.
- It induces bud and seed dormancy.
- It promotes the abscission of leaves, flowers and fruits by forming abscission layers.
- ABA plays an important role in plants during water stress and during drought conditions. It results in loss of turgor and closure of stomata.
- In Cannabis sativa, induces male flower formation on female plants.
- It promotes sprouting in storage organs like Potato.
- It inhibits the shoot growth and promotes growth of root system. This character protect the plants from water stress. Hence, ABA is called as **stress hormone.**

15.3 Plant Movements



Plants have the capacity for changing their positions in response to external or internal stimuli, which are known as **plant movements**. Movements are basically of two types: I. Vital movements and II. Physical movements (hygroscopic) (Figure 15.10).

I. Vital movements

Vital movements are those which are exhibited by the living cells or plants or

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